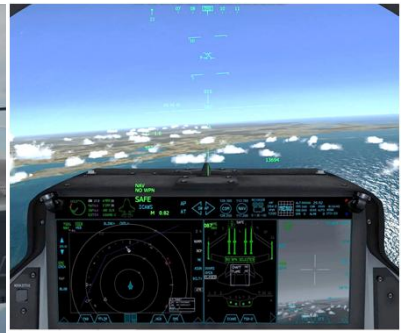
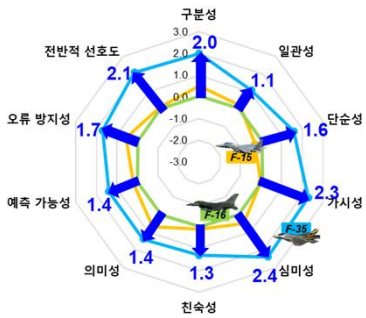
Vision &
Visual Display Design





A study of LAD/HUD PVI design methodology considering human factors (You et al., June 2016 ~ June 2017; sponsored by Korean Aerospace Industry)

Contents

Part I

- Visual Capabilities
- Hardcopy Text Design
- VDT Screen Design
- Graphics, Symbols, and Codes

Part II

- Dynamic & Quantitative Displays
- Object Displays & Integration
- Qualitative Displays & Signals
- Representational & Advanced Displays
- Quiz

Learning Goals (1/2)

GOALS

- Identify the major anatomical structures of the eye (**Cornea, Lens, Retina, and Fovea**) in vision.
- Distinguish between the anatomical functions of **Rods and Cones** regarding **daylight/night vision, color vision, motion perception, and dark/light adaptation**.
- Explain **visual accommodation** and **dark focus** as well as the optical mechanisms of **Myopia** and **Hyperopia**.
- Distinguish **Visual acuity** from **Contrast Sensitivity**.
- Compute **visual angle** and evaluate how environmental and personal factors (**luminance, contrast, exposure time, target motion, age, and training**) **affect visual acuity and contrast sensitivity**.

- Differentiate among **visibility, legibility, and readability**, and identify the typography variables that influence each.
- Calculate the required **stroke width** and **letter height** for printed text based on viewing distance and Snellen acuity, and the **irradiation effect** under varying illumination levels.

Learning Goals (2/2)

GOALS

- Identify the **technological and perceptual factors**, such as lower image resolution and spatial frequency channel stimulation, that cause **slower reading speeds on VDT screens** compared to hardcopy.
- Select **appropriate design features for VDTs**, including character matrices, display polarities, color, screen density, grouping, and high-validity highlighting techniques, while avoiding disruptive blinking text.
- Design **graphics, symbols, and visual codes** for better **accuracy, speed, ease of recall, and preference**.
- Correct **perceptual distortions** in graphical representations to ensure accurate quantitative perception.

- Apply **perceptual principles**, such as **figure-to-ground articulation, figure boundaries, closure, simplicity, and unity,** to design unambiguous symbols that are processed faster than verbal signs.
- Assign **appropriate visual coding dimensions to specific user tasks,** prioritizing color coding for searching tasks and alphanumeric coding for identification tasks.

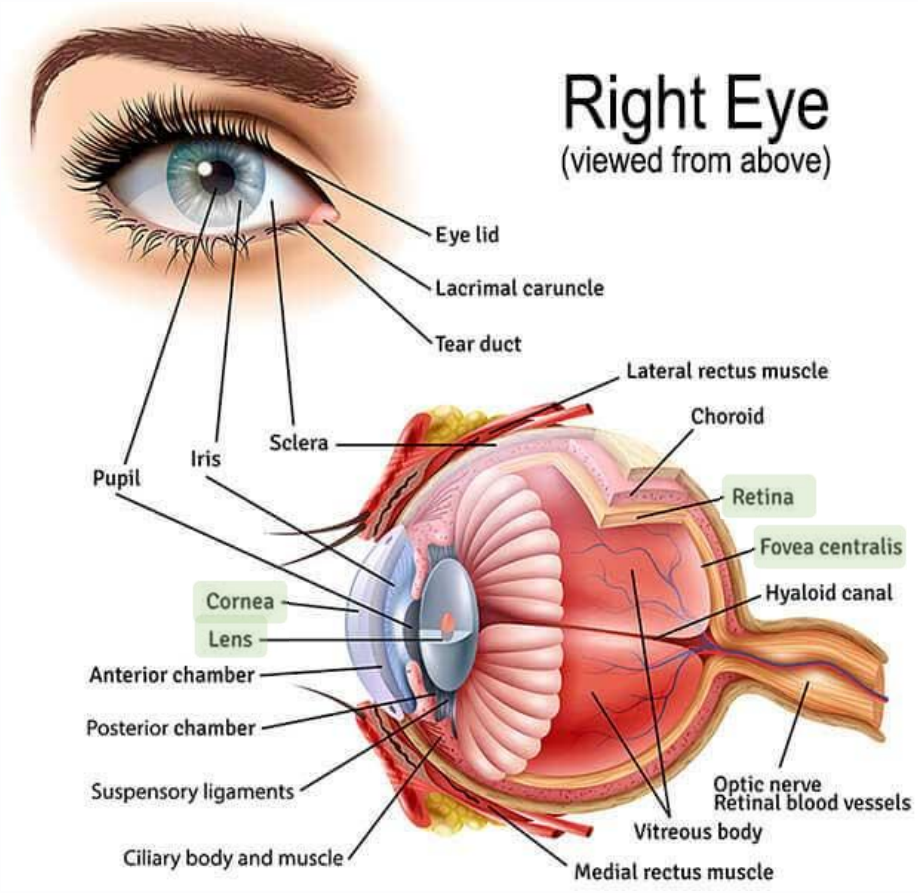
Section 01

Visual Capabilities

Understanding the anatomy, physiology, and perceptual limits of the human eye for effective display design

The Process of Seeing: Eye Anatomy

The eye acts like a camera: **Light** passes through the **cornea** and **lens** to focus an inverted image on the **retina**.



Cornea

Transparent outer layer. Provides 65-75% of the eye's focusing power. Light enters here first.

Lens (Accommodation)

Adjustable structure that fine-tunes focus. Changes shape (accommodation) to focus on near vs. far objects.

Retina

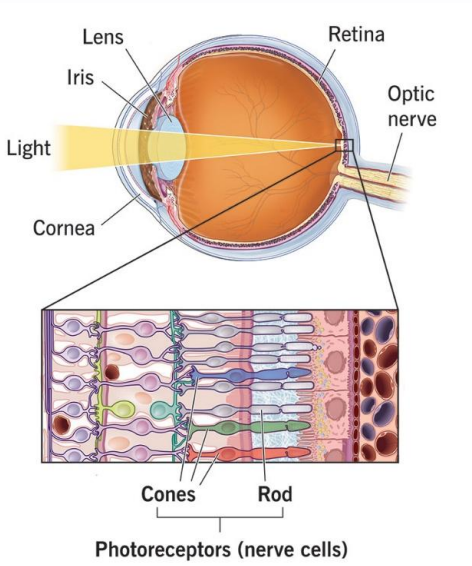
Light-sensitive lining at the back. Contains **Rods** (night vision) and **Cones** (color/daylight).

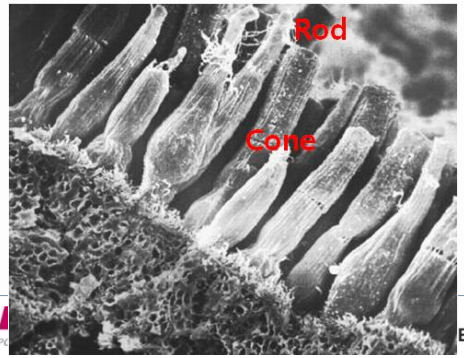
Fovea

Small central pit composed **exclusively of cones**. Responsible for **sharp central vision and highest visual acuity**.

Types of Photoreceptors: Rods and Cones

Under dim light conditions (e.g., at night), the eyes rely almost entirely on **Rods** — meaning **color perception is effectively lost**!





Cones

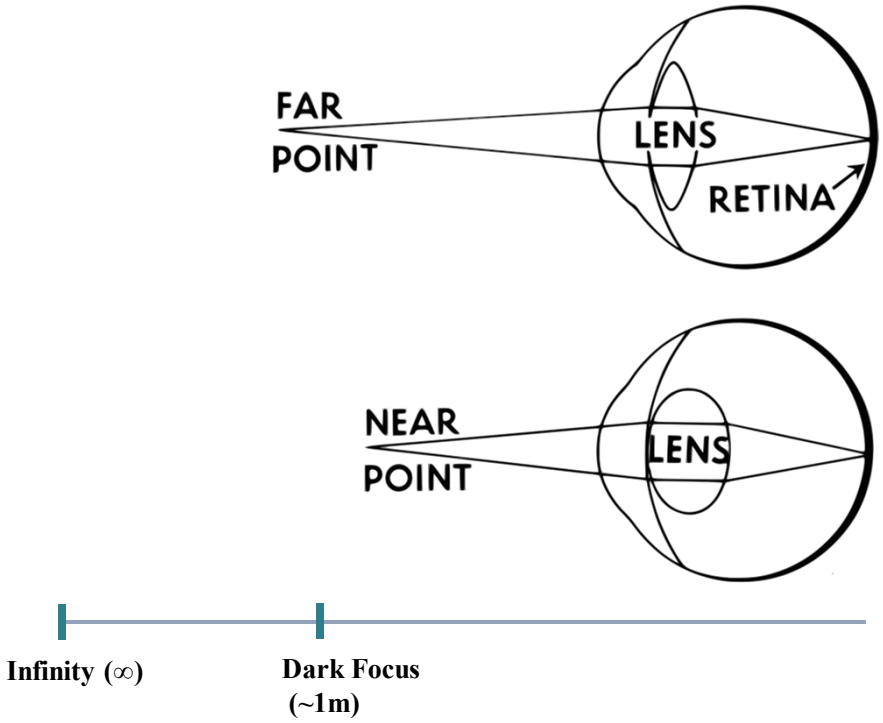
Quantity	~ 6 to 7 Million
Location	Concentrated in the Fovea (Center)
Function	Daylight vision (Photopic)
Provides	Color Vision High Acuity (Detail)

Rods

Quantity	~ 130 Million
Location	Peripheral Retina (Outside Fovea)
Function	Low-light vision (Scotopic)
Provides	Black & White Only Motion Detection

Visual Accommodation & Dark Focus

Accommodation is the eye's active process of adjusting focus for different distances. In contrast, **dark focus** is the eye's natural intermediate resting state in the absence of visual stimuli, often leading to **night myopia**.



Accommodation

The process where the lens changes shape to focus light on the retina. The lens thins for distant objects and thickens for near ones .

Near Point & Far Point

Near Point: The closest distance the eye can focus on.
Far Point: The farthest distance (normally infinity).

Dark Focus (Resting State)

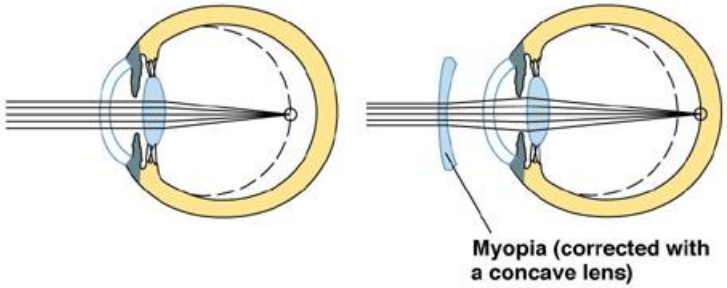
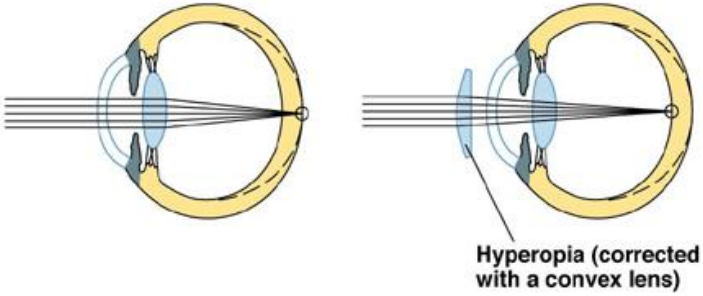
In total darkness, the eye does NOT focus at infinity. Instead, it spontaneously reverts to a resting distance of about **1 meter (1 Diopter)**.

⚠ Design Implication: HUDs

In empty visual fields (like darkness or looking at a Head-Up Display), eyes tend to accommodate to their dark focus (~1m) rather than optical infinity. This causes pilots to **misjudge size and distance** of real-world objects.

Myopia vs. Hyperopia

Myopia (nearsightedness) makes **distant objects blurry** as light focuses in front of the retina. **Hyperopia (farsightedness)** makes **near objects blurry** as light focuses behind the retina. As people age, they often develop **presbyopia**, a condition where the eye's natural **lens loses its flexibility**, making it **difficult to focus on close objects**.

Type	Focal Point	Illustration
Myopia (Near-sightedness)	In front of the retina	
Hyperopia (Far-sightedness)	Behind the retina	

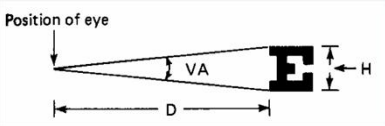
Visual Acuity (VA)

Definition

Visual Acuity is the ability of the eye to **discriminate fine detail** and depends on the **accommodation capability of the eyes**.

Visual Angle Formula

$$VA = \frac{H}{D} \times 3438$$

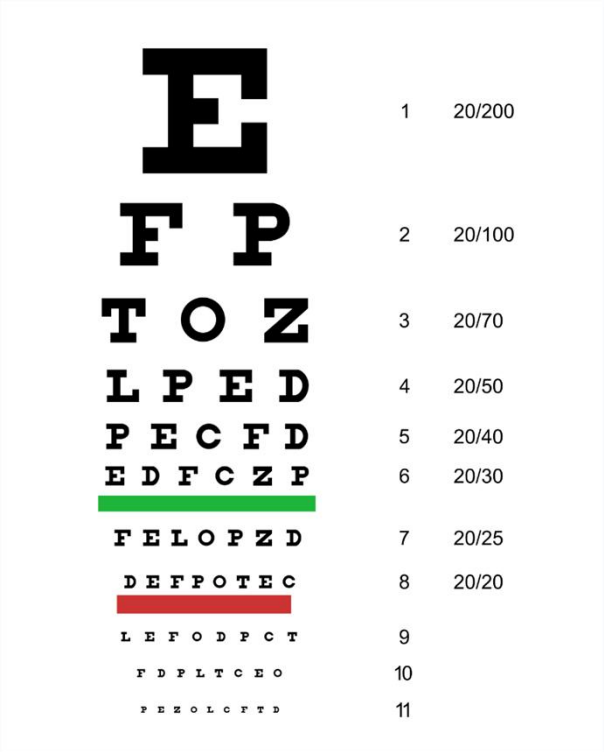


VA	H	D	$3,438$
Visual Angle (minutes of arc)	Height of Target	Viewing Distance	Constant (radians to min)

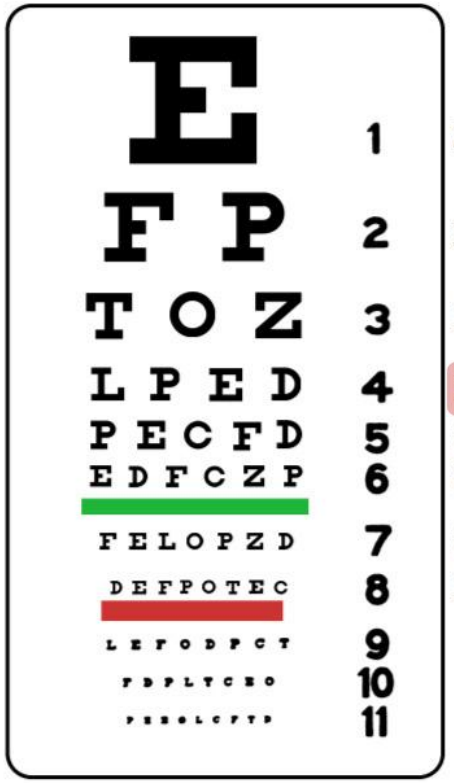
- **Standard Normal Vision (20/20):** Ability to resolve a target subtending **1 minute of arc at 20 feet (6 m)**.
- **Measurement Tools:** Snellen eye chart (most common) or Landolt C rings.

Snellen Chart

Standard clinical tool for measuring acuity



Snellen Acuity Interpretation



Examinee

Normal

20/200 = 0.1

20/100 = 0.2

20/70

20/50 = 0.4

20/40 = 0.5

20/30

20/25

20/20 = 1.0

Examinee can read at 20 ft (6 m)
what a **person with normal vision can read at 50 ft (15 m)**

Contrast Sensitivity (CS)

Definition

Contrast sensitivity is the ability of the eye to **distinguish an object from its background, particularly in low-light, foggy, or low-contrast conditions**. Unlike visual acuity (clarity), contrast sensitivity measures the eye's ability to identify subtle differences in shades and patterns.

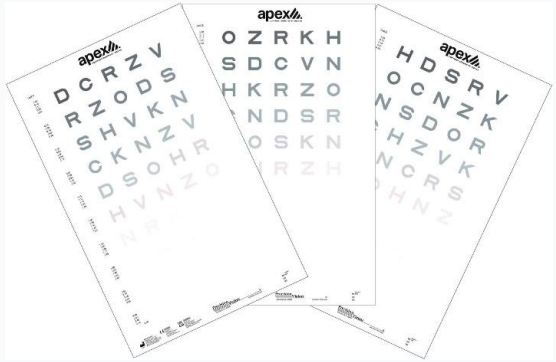
Key Aspects

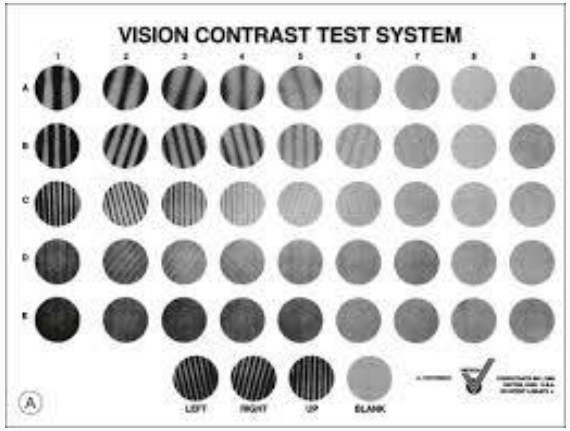
- **Importance in Daily Life**: It is crucial for driving at night, navigating stairs, reading, and detecting objects that lack clear outlines, such as detecting a grey curb against dark tarmac.
- **Low Contrast Sensitivity vs. Visual Acuity**: It is possible to have excellent 20/20 vision (sharpness) but still have poor contrast sensitivity, resulting in hazy or blurry vision in low-contrast environments.
- **Causes of Reduced Contrast Sensitivity**: Reduced contrast sensitivity can be a **symptom of conditions like cataracts, glaucoma, macular degeneration, or normal age-related changes**.

- **Measurement Tools:** Specialized charts, such as the Pelli-Robson test, which displays letters that gradually fade from black to gray.

Contrast Sensitivity Charts

Standard clinical tool for measuring contrast sensitivity





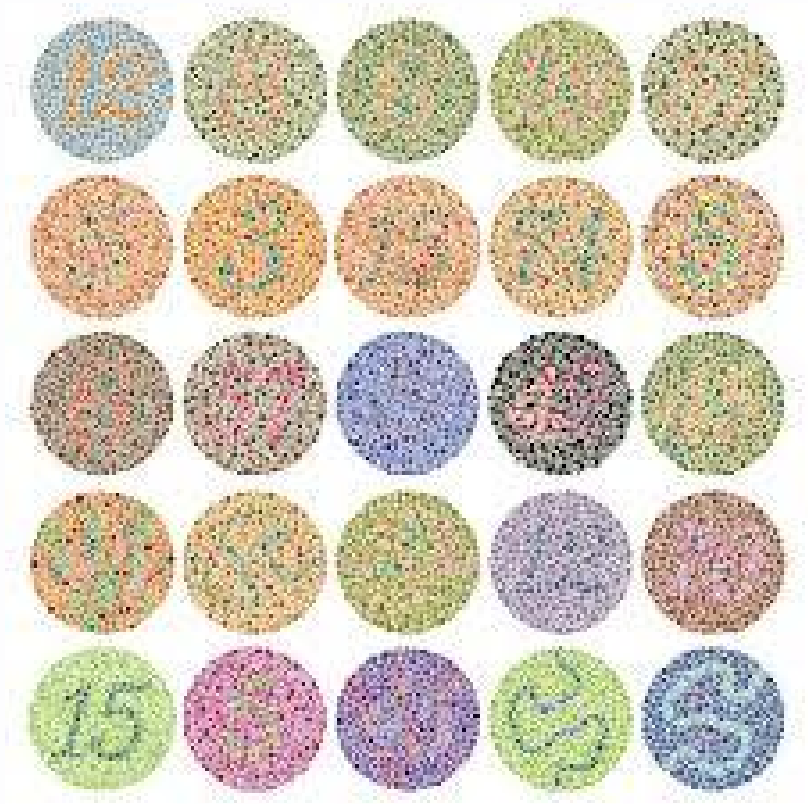
Factors Affecting VA & CV

Factor	Effect on Visual Acuity & Contrast Sensitivity
Luminance Level ⇑	Visual acuity and contrast sensitivity increase with higher levels of light or background luminance and then level off. This improvement occurs because higher light levels activate the cones.
Contrast ⇑	When the contrast between a visual target and its background is low, the target must be larger to be equally discriminable compared to a target with greater contrast.
Exposure Time ⇑	Under high illumination conditions, visual acuity improves as exposure time **increases up to 100 or 200 ms, after which it levels off**.
Target Motion ⇓	The movement of a target or the observer decreases visual acuity, a capability known as dynamic visual acuity. Acuity deteriorates rapidly when **the rate of motion exceeds approximately 60 degrees per second**.
Age ⇓	Both visual acuity and contrast sensitivity decline continuously with age. This decline generally begins **after age 40**.
Training ⇑	Evidence suggests that training and practice can improve both static and dynamic visual acuity as well as contrast sensitivity.

Color Vision

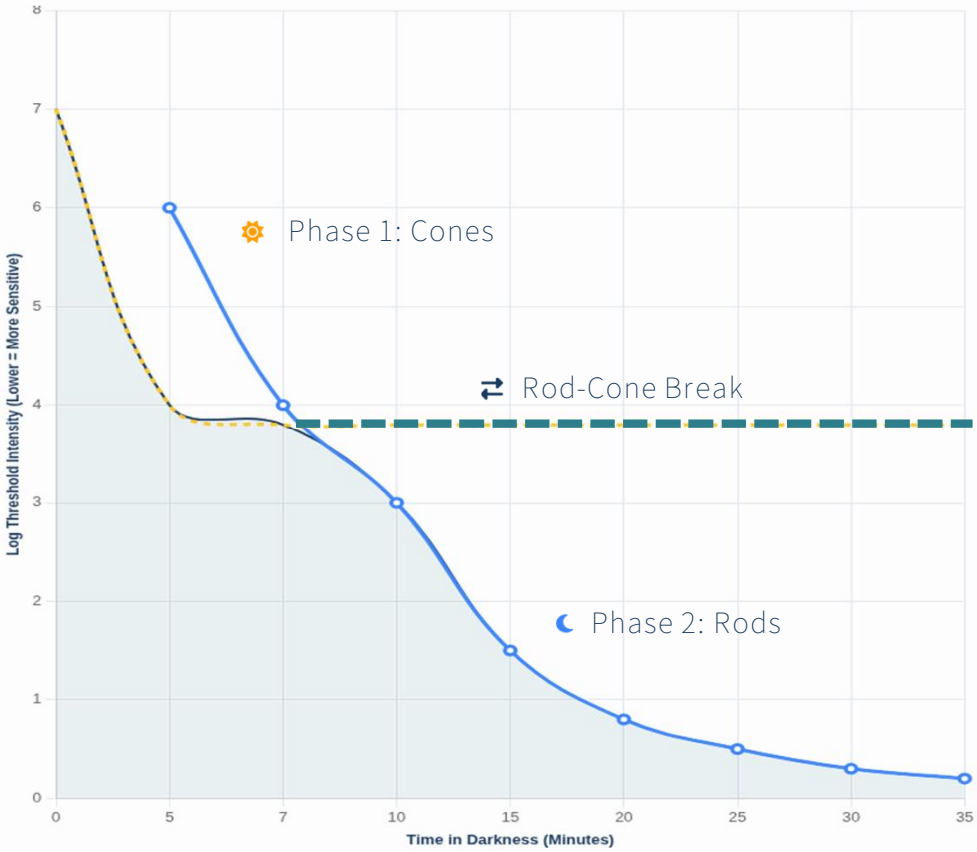
Specialized cones enable color perception by sensing light. **Deficiency in color perception occurs when certain cone types—L, M, or S—fail**, hindering a person's capacity to discriminate between specific color ranges.

3 Cone Types	Red Sensitive (L-cones) Green Sensitive (M-cones) Blue Sensitive (S-cones)
Deficiency Rates	**Males:** ~8% affected **Females:** ~0.5% affected
Common Types	**Dichromats (Severe color vision deficiency):** Missing one cone type **Anomalous Trichromats (Mild color vision deficiency):** One cone type has shifted sensitivity



Dark Adaptation Process

Dark adaptation **increases light sensitivity in the dark.** It involves **rhodopsin regeneration**, taking 20–30 minutes. Factors like **aging** and **vitamin A deficiency** can slow this process.



Phase 1: Cone Adaptation

Occurs rapidly during the first **0–5 minutes**. Sensitivity increases but levels off quickly. Limited to photopic (daylight) vision levels.

Rod-Cone Break

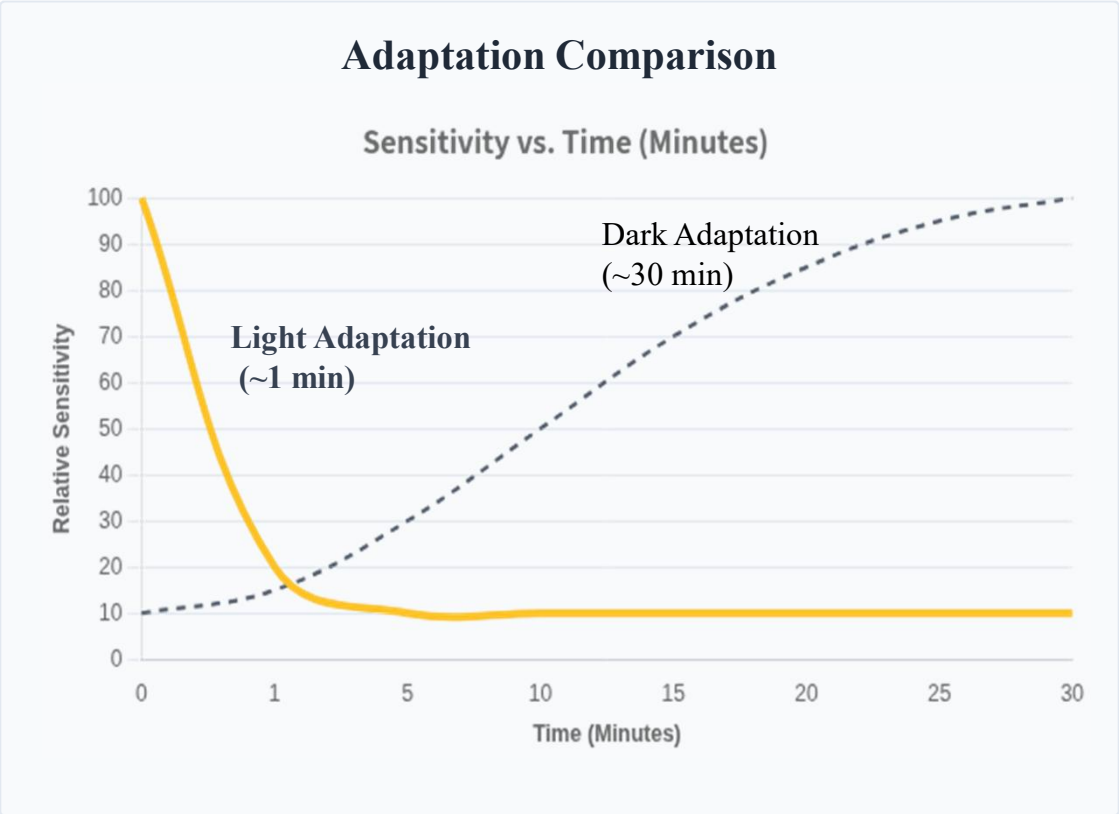
Around **5–10 minutes**, the curve dips sharply as rods become more sensitive than cones and take over the adaptation process.

Phase 2: Rod Adaptation

Continues from **5–35 minutes**. Rods provide a massive increase in sensitivity (scotopic vision), allowing vision in very dim light.

Light Adaptation Process

Light adaptation rapidly **lowers light sensitivity** via **pupil constriction** and **rod-to-cone switching**. It's faster than dark adaptation, finishing in less than one min. as cones recover quickly.



Rapid Process

When moving from dark to bright light, the eye adapts very quickly. The entire process takes only about **1 minute** to complete significantly.

Sensitivity Decrease

Light adaptation involves a rapid **decrease in sensitivity**. The rod pigments bleach out instantly, and cones take over vision responsibilities.

Mechanism

Neural adaptation happens much faster than chemical regeneration. **The pupil also constricts rapidly to limit light entry** (pupillary reflex).

Section 02

Design Principles for Hardcopy Text

Exploring visibility, legibility, and readability criteria for optimal text design in printed materials

HF/E Criteria for Text Design

Visibility → Legibility → Readability

Criteria	Definition	Design Variables
Visibility (detectability)	The quality of a character or symbol that makes it **separately visible from its surroundings**.	- Font size
Legibility (discriminability)	The quality of alphanumeric characters that makes it possible for each one to be **identifiable from others**.	- Stroke width-to-height ratio - Width-to-height ratio - Typeface - Contrast
Readability (ease of understanding)	The quality of an information content that makes it **recognizable in continuous text**.	- Inter-letter spacing - Inter-word spacing - Inter-line spacing - Margins - Sentence structure

Typography: Stroke Width-to-Height Ratio

Condition	Illumination	Recommended Ratio
A Black on White	Normal Light	**1:6 to 1:8**
A White on Black	Normal Light	**1:8 to 1:10**
B Boldface	Low Illumination	**1:5**
Back-lit (Transilluminated)	Dark Environment	**1:12 to 1:20**

Black On White (1:6)

HFES 2026

White On Back (1:10)

HFES 2026

Interpretation

A ratio of **1:6** means the stroke width is 1/6th of the letter height. Higher second numbers (e.g., 1:10) indicate **thinner** strokes.

Irradiation Effect

White-on-black text appears to **"Spread"** or glow into the surrounding black background due to irradiation. Therefore, it requires a **thinner stroke width** (higher ratio) to maintain legibility compared to black-on-white text.

Calculating Stroke Width & Letter Height

Step 1: Calculate Stroke Width

$$W_s = 1.45 \times 10^{-5} \times S \times D$$

W_s	S	D
Stroke Width (inches)	Denominator of Snellen Acuity (e.g., 20/40 → 40)	Viewing Distance (inches)

Step 2: Calculate Letter Height

$$H_L = W_s/R$$

H_L	W_s	R
Letter Height (inches)	Stroke Width (from Step 1)	W_s:H_L Ratio (e.g., 1:5 → 0.2)

Example

Viewing Distance (D)	100 ft (1200 in)
Snellen Acuity (S)	20/40 (S = 40)
W_s:H_L Ratio (R)	1:6 (0.167)

Calculation:

1. $W_s = 1.45 \times 10^{-5} \times 40 \times 1200$

   W_s = **0.696 in**

2. $H_L = 0.696 / 0.167$

   H_L = **4.17 in**

Letter Case & Text Spacing

Letter Case

- **Lowercase text** is generally **easier to read** than all uppercase.
- **Ascenders and descenders** create a distinctive "**word envelope**"
- **Uppercase words** have a rectangular block shape, making them **harder to distinguish**.

UNIVERSITY	University
Rectangular Shape (Harder)	Distinctive Envelope (Easier)

Spacing Guidelines

- **Interletter Spacing: Close-set type** allows **more characters per fixation**, increasing **reading speed**
- **Interline Spacing: Adequate leading** (vertical space) **reduces eye strain and helps track lines**
- **Point Size:** 9–11 pt is standard for 12–16 inch reading distance

Spacing Impact

- Too tight line spacing makes text feel dense.
- **Optimal line spacing** improves readability and reduces visual tracking errors.

Example: Interletter Spacing

Design A

Regular spacing of text type (regular density)

The ESS Performance Series is both a choice and a statement. The choice is to continue ESS's long tradition of excellence by trimming costs without

Design B

Close-set text type (high density)

The ESS Performance Series is both a choice and a statement. The choice is to continue ESS's long tradition of excellence by trimming costs without sacrificing performance and by omitting

Which has better readability?

Example: Interline Spacing

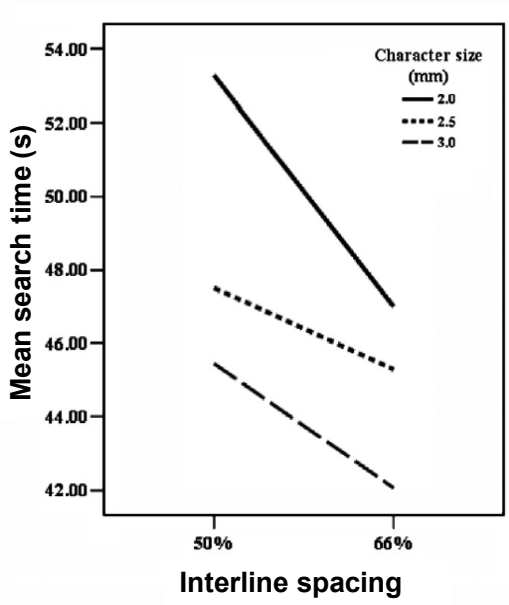
Design A

Adding graphical marks helps to show the relationship between the primary and supporting material when they cannot be adjacent. The margin callout technique avoids altering the primary material in order to display supporting material. Instead, it places the supporting material in existing white space on the page, for example, the margins.

Design B

Adding graphical marks helps to show the relationship between the primary and supporting material when they cannot be adjacent. The margin callout technique avoids altering the primary material in order to display supporting material. Instead, it places the supporting material in existing white space on the page, for example, the margins.

Which has better readability?



Measuring Readability: Flesch Reading Ease Score

THE FORMULA

$$\text{Score} = 206.835 - (0.846 \times S) - (1.015 \times W)$$

S = Total syllables per 100 words

W = Average number of words per sentence

How to Improve Readability

- Use **active voice** instead of passive voice
- Keep sentences **short and direct**
- Prefer **shorter, simpler words** over complex vocabulary

Score Range	Difficulty Level	Typical Example
90 – 100	Very Easy	Comics, Children's Books (5th grade)
80 – 90	Easy	Pulp Fiction, Conversational English
70 – 80	Fairly Easy	Slick Fiction, Informal English
60 – 70	Standard	Digest Magazines, Plain English
50 – 60	Fairly Difficult	Quality Magazines (Time, Newsweek)
30 – 50	Difficult	Academic Journals, College Textbooks
0 – 30	Very Difficult	Scientific Papers, Legal Documents

Higher scores indicate easier reading. A score of 60-70 is generally considered acceptable for standard adult reading materials.

Section 03

Designing Text for VDT Screens

Optimizing readability, performance, and user comfort on visual display terminals

Why is VDT Reading Slower?

20-30%

Slower Reading Speed

compared to hardcopy text

> "People proofread about 20 to 30 percent slower when the material was presented on a VDT than when it was presented on hardcopy."
>
> — Gould & Grischkowsky (1984)

Underlying Causes

Lower Image Quality (Resolution)

Pixel-based characters on computer screens lack the sharp edges of printed text, causing the eye's accommodation mechanism to "hunt" for focus.

Spatial Frequency Channels

Dot-matrix characters stimulate specific spatial frequency responders in the brain differently than solid strokes, potentially reducing processing efficiency.

Design Solution

→ **Use higher resolution screens to reduce the performance gap closer to hardcopy levels.**

Character Matrix & Polarity

Resolution

Character Matrix Size

- **Minimum Standard:** 7 × 9 dot matrix is required for continuous text reading
- **Enhanced Legibility:** 15 × 24 matrix (or higher) preferred for extensive tasks
- Higher resolution makes individual pixels less distinct, aiding accommodation

Visual Comparison

E
Low Resolution (7×9)
Visible Pixels

→

E
High Resolution (15×24)
Paper-like

Contrast

Display Polarity

- **Positive Polarity** (Dark-on-Light): Generally preferred; similar to paper
- **Negative Polarity** (Light-on-Dark): Reduces flicker perception but increases glare risk
- Choice often depends on ambient lighting conditions

Polarity Modes

Positive
Preferred

Negative
Low Flicker

Color Guidelines for VDT Screens

AVOID

Saturated **Red** and **Blue** combinations, especially on dark backgrounds.

Chromostereopsis Effect

RED TEXT

BLUE TEXT

Appears Closer

Red appears closer than blue due to chromatic aberration in the eye, causing focusing difficulty and fatigue.

Avoid Saturated Color Pairs

Do not place saturated red/blue or red/green next to each other. The eye cannot focus on both simultaneously without strain.

Watch Background Contrast

Avoid using blue text on dark backgrounds (low acuity for blue). Avoid red text on green backgrounds (color deficiency issues).

Limit Color Palette

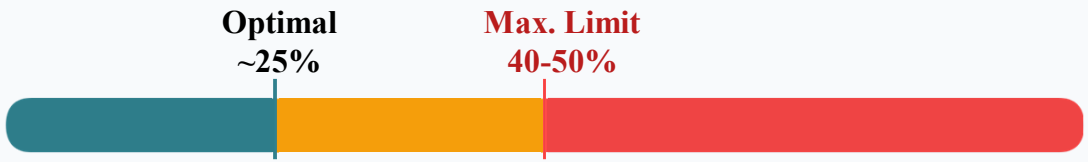
Restrict usage to **5 to 7 colors** per display. Too many colors create visual clutter and increase search time.

Code for Meaning

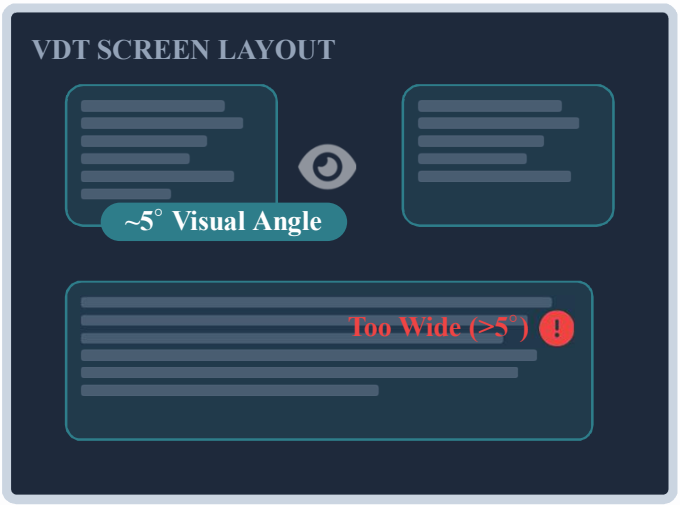
Use color to code categories or status (e.g., Red = Danger), not just for decoration. Ensure redundant coding (e.g., shape + color) for color-blind users.

Screen Density & Grouping

Overall Density Guidelines



Tullis (1984): Search time and errors increase significantly above 40-50% density



High density crowds information, while very low density increases scan distance. Balance is the key!

Overall Density

Percentage of available character spaces used. Keep below 40-50%. Optimal is around 25% for best search performance.

Grouping Strategy

Organize information into perceptual groups that subtend approximately **5 degrees** of visual angle.

Group Size Metrics

Ideally ~12-14 characters wide by 6-7 lines high. This matches the foveal span for efficient processing.

Layout Complexity

Use **columns** instead of running text. Predictable vertical alignment reduces eye movement and search time.

Highlighting & Blinking

Technique	Use When	Avoid When
Highlighting	Important items Validity > 50%	Overused (clutter) Low validity
Blinking	Critical warnings Emergency only	Routine information Distraction risk

Color Highlight — Recommended

Status: Processing complete.

Blinking Text — Use Sparingly

System: CRITICAL FAILURE

Validity of Highlighting

Highlighting is only effective if it has **high validity**. If users learn that highlighted items are often irrelevant (low validity), they will eventually ignore the highlighting cue entirely.

Blinking Distraction

Blinking text is extremely **hard to read** and draws attention away from other tasks. It should be reserved exclusively for **urgent messages** that require immediate action.

Section 04

Graphics, Symbols, & Visual Codes

Perceptual principles for effective graphic design, symbolic representation, and coding systems

HF/E Criteria for Graphics, Symbols & Visual Codes

Criteria	Definition
Accuracy **(Ease of recognition)**	- % correct matching (% error) - Level of correct interpretation
Speed	- Reaction time
Ease of recall	- % correct recall - Likert or rating scale (5-pont, 7-point, 9-point scales)
Preference	- Likert or rating scale (5-pont, 7-point, 9-point scales)

Symbols of mandatory-action messages used in a study of recognition and recall of such symbols. The percentages below the symbols are the percentages of correct recognition, as follows: O = original test; R = recall 1 week later. (*Source: Adapted from Cairney and Siess, 1982, Fig. 1.*)

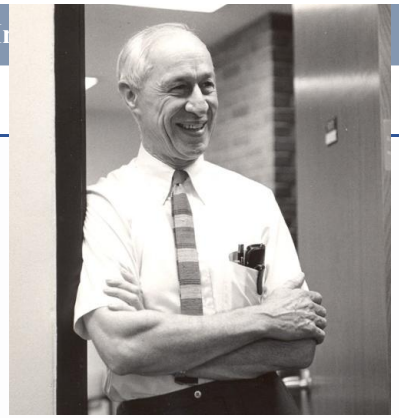
	1: Must use ear protection	2: Must use eye protection	3: Must use foot protection	4: Must use hand protection	5: Must use head protection	6: Must use breathing protection
Original test (O)	10%	20%	30%	50%	37%	13%
Recall 1 week later (R)	73	96	100	97	97	97

Original test

Recall 1 week later

Likert Scale

A **Likert scale** is a psychometric tool used in surveys to **measure attitudes, opinions, or behaviors** by asking respondents to rate their **level of agreement with a statement**. **Typically 5 or 7 points**, it ranges from one extreme (e.g., **Strongly Disagree**) to another (e.g., **Strongly Agree**), **converting qualitative opinions into quantitative data**. Likert scales are **a specific type of rating scale used to measure attitudes or agreement**.



Rensis Likert
(1903 ~ 1981)
an American organizational and social psychologist;
Earned a Ph.D. in Psychology from Columbia University

Please circle the number that represents how you feel about the computer software you have been using

I am satisfied with it
Strongly Disagree ---1---2---3---4---5---6---7--- Strongly Agree

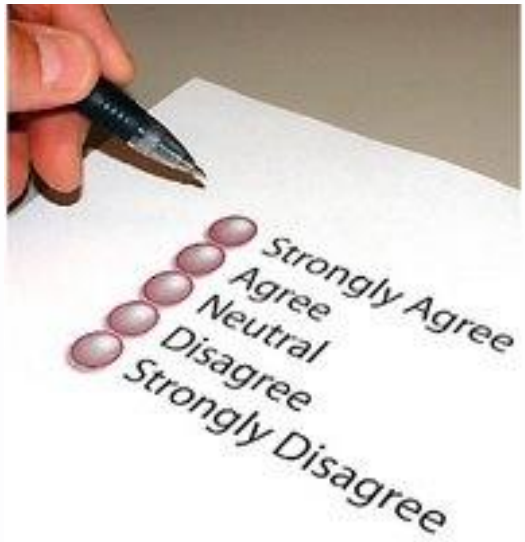
It is simple to use
Strongly Disagree ---1---2---3---4---5---6---7--- Strongly Agree

It is fun to use
Strongly Disagree ---1---2---3---4---5---6---7--- Strongly Agree

It does everything I would expect it to do
Strongly Disagree ---1---2---3---4---5---6---7--- Strongly Agree

I don't notice any inconsistencies as I use it
Strongly Disagree ---1---2---3---4---5---6---7--- Strongly Agree

It is very user friendly
Strongly Disagree ---1---2---3---4---5---6---7--- Strongly Agree



Rating Scale

A rating scale is a closed-ended survey tool used to **measure attitudes, opinions, or performance by assigning numerical or descriptive values** (e.g., 1–5, Poor to Excellent) to a subject.

- **Likert Scale (Agreement):** Measures agreement with a statement.
  Strongly Disagree / Disagree / Neutral / Agree / Strongly Agree
- **Quality Scale:** Evaluates performance or satisfaction.
  Very Poor / Poor / Fair / Good / Very Good
- **Satisfaction Scale:** Measures customer experience.
  Very Dissatisfied / Dissatisfied / Neutral / Satisfied / Very Satisfied
- **Frequency Scale:** Measures how often something occurs.
  Never / Rarely / Sometimes / Often / Always
- **Numerical Scale:** Uses numbers, commonly 1-5 or 1-10.
  Example: "Rate your experience from 1 (lowest) to 5 (highest)."

Quality Evaluation Questions

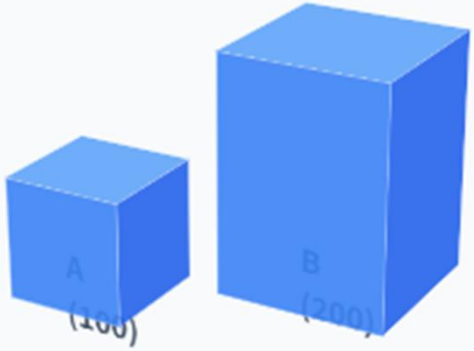
Questions	Very Poor	Poor	Fair	Good	Very Good
How useful is it for information search?	①	②	③	④	⑤
How easy is it to use?	①	②	③	④	⑤
How easy is it to learn?	①	②	③	④	⑤
How explicit is the organization?	①	②	③	④	⑤
How clear is the terminology?	①	②	③	④	⑤
Questions	**Very Dissatisfied**	**Dissatisfied**	**Somewhat Satisfied**	**Satisfied**	**Very Satisfied**
What is your overall satisfaction?	①	②	③	④	⑤

Avoid Perceptual Distortions in Data Graphs (1/3)

Design Rule: "Use the **Simplest** representation that communicates your data **accurately without decorative distortion**."

Poor Design

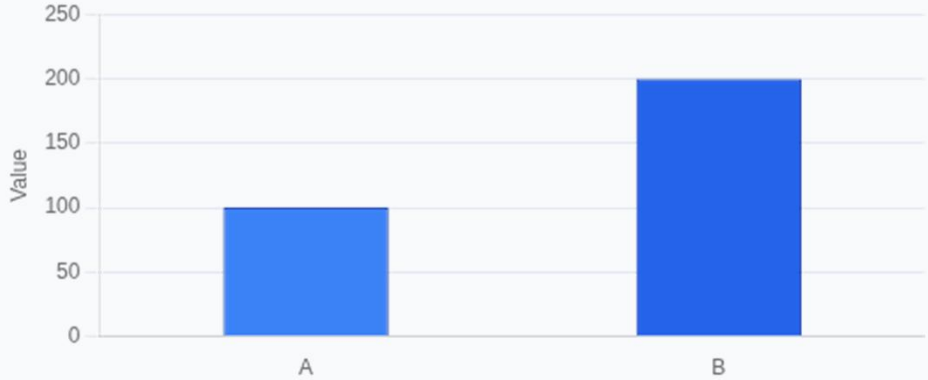
3D Distortion



Volume Illusion: Users perceive the **VOLUME** of the 3D block rather than just the height. Block B represents 2x the value of A, but appears visually as **~4x larger** due to added depth and width.

Good Design

2D Accuracy

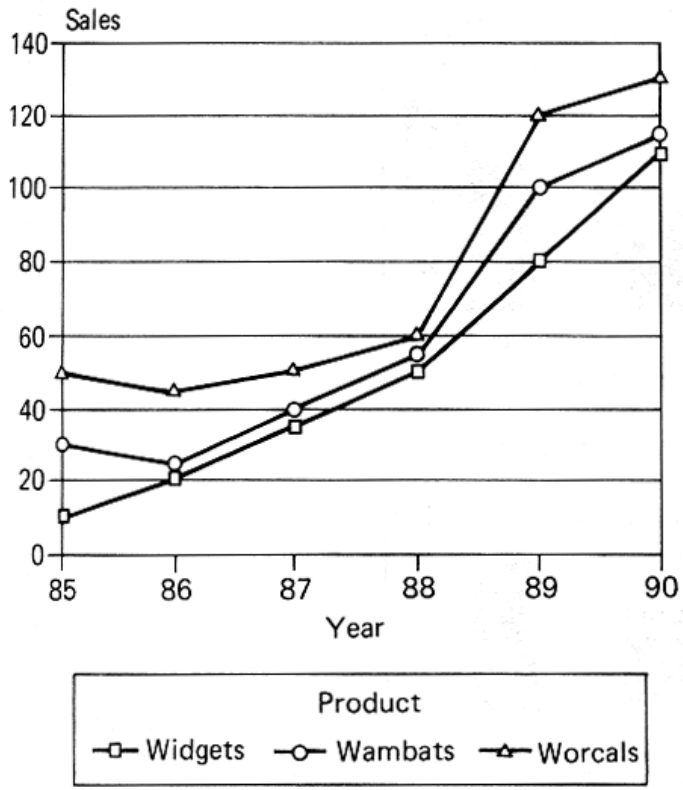


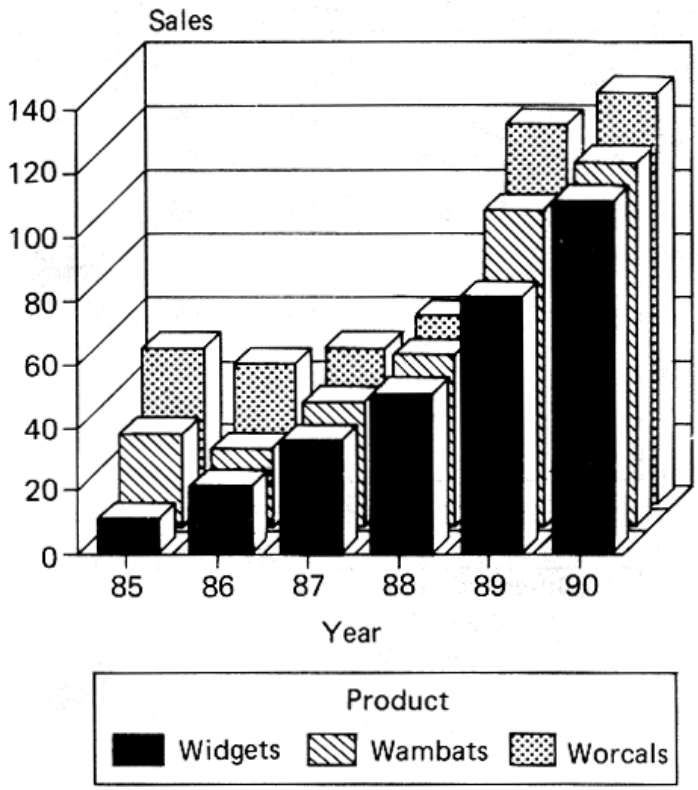
Simple 2D Representation: Height is the **only dimension** that changes. This maps the visual variable directly to the data value (1:1 ratio), ensuring accurate perception and comparison.

Avoid Perceptual Distortions in Data Graphs (2/3)

Poor Design **Good Design**

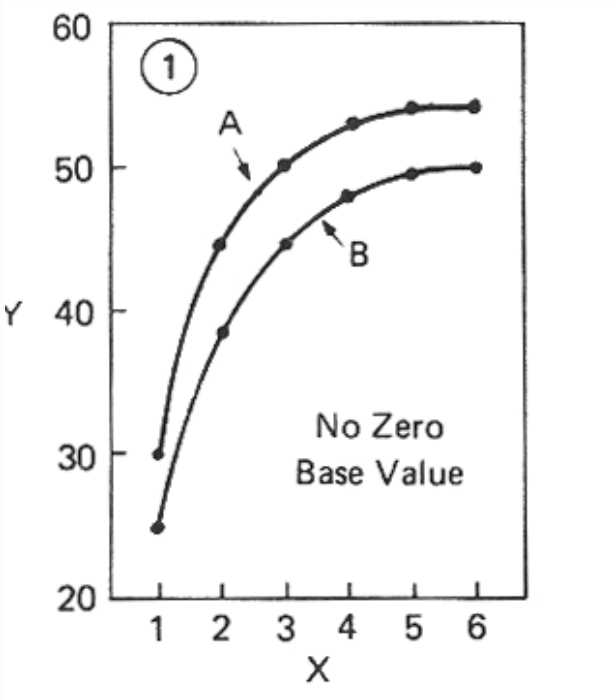
Illustrations of two graphic representations of data. Left, multiple-line graph; right, three-dimensional clustered bar chart. The data presented is the same in both representations.



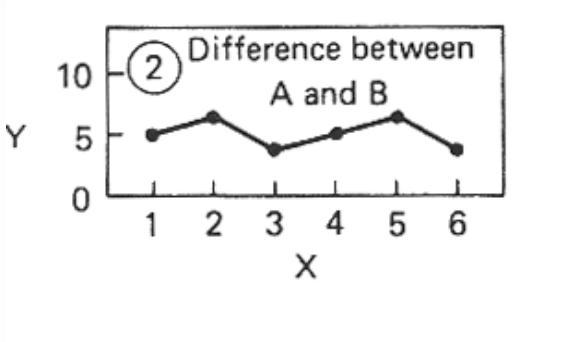


Avoid Perceptual Distortions in Data Graphs (3/3)

Poor Design



Good Design



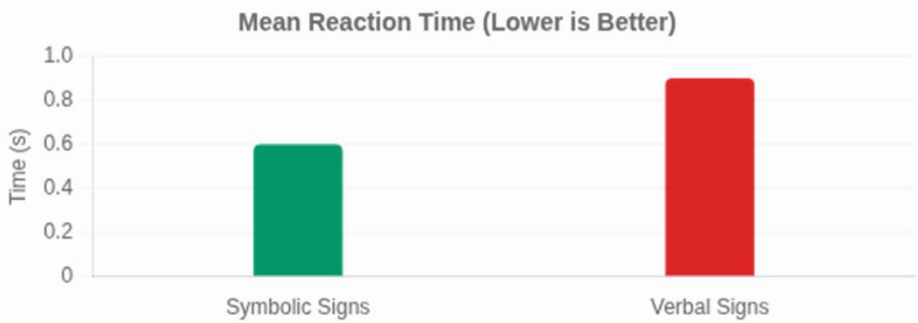
Symbols vs. Verbal Signs

~0.6s

Reaction Time (Symbolic)

vs. ~0.9s for Verbal Signs

> *"Symbolic signs showed systemic superiority in reaction times over verbal signs, particularly under visually degraded conditions."*
>
> — Ellis & Dewar (1979)



Visual Comparison (Degraded Condition)

Symbolic Sign

Recognized ✓

Shape remains distinct

Verbal Sign

Unreadable ✗

Text blurs together

Why Are Symbols Faster?

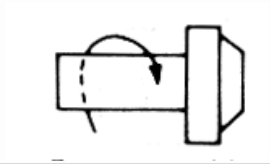
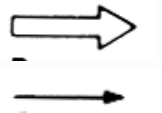
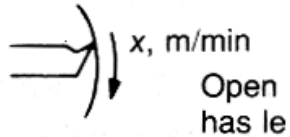
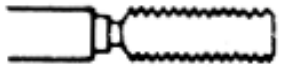
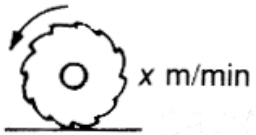
No Language Recoding

Symbols allow for direct visual recognition without the intermediate step of translating text into meaning (recoding), saving cognitive processing time.

Superior Legibility Distance

Simple geometric shapes and standardized icons can be resolved at greater distances and under poorer viewing conditions (fog, rain, speed) than complex letterforms.

Perceptual Principles for Symbols

Perceptual Principle	Description	Poor Design	Good Design
Figure-to-Ground	**Clear and stable articulation** is essential. The figure must stand out distinctly from its background context.		
Figure Boundaries	A **solid shape** (contrast boundary) is preferable to a line boundary. Solid forms are processed faster.		
Closure	**Closed figures** are easier to process than open or discontinuous ones. The eye naturally seeks to close gaps.		
Simplicity	Symbols should be as **simple** as possible. Include only **essential features** necessary for recognition.		
Unity	Symbols should be as **unified** as possible. Elements should form a **cohesive, integrated whole**.		

Visual Coding Dimensions: Which and When?

Key Principle: Match the Visual Code to the Task

Use **Color codes** when the user needs to quickly ***scan and locate*** items (Searching).
Use **Alphanumeric codes** when the user needs to **identify and name** specific items (Identification).

Coding Type	# Levels	Best Used For	Caution / Limitation
Color	9–24	Searching	Color-blind users (~8% males)
Shape / Symbol	15+	Both Search & ID	Complexity increases processing time
Alphanumeric	Unlimited	Identification	Requires literacy; poor for search
Flash Rate	2	Urgency / Warning	Annoyance; can be distracting
Inclination Angle	24	Quantitative Estimation	Low accuracy; potential confusion